

Summary and Recommendation

1. Project Objective

The notebook performs an **Exploratory Data Analysis (EDA)** on a customer churn dataset to identify patterns and factors associated with customer attrition.

2. Dataset Overview

Metric	Value
Total Records	7,043 customers
Total Features	21 columns
Missing Values	0
Duplicate Records	0
Duplicate Customer IDs	0

Main Features

- Demographics: Gender, SeniorCitizen, Partner, Dependents
- Account Information: Tenure, Contract Type
- Services: PhoneService, InternetService, OnlineSecurity, TechSupport, Streaming Services, etc.
- Billing: MonthlyCharges, TotalCharges, PaymentMethod
- Target Variable: **Churn**

3. Data Cleaning Performed

TotalCharges Fix

- Blank values in **TotalCharges** were replaced with **0**
- Column converted from string to **float**

SeniorCitizen Transformation

- Converted:
 - **1** → **Yes**
 - **0** → **No**

4. Statistical Summary

Numerical Features

Feature	Mean	Median	Min	Max
Tenure	32.37 months	29	0	72
Monthly Charges	64.76	70.35	18.25	118.75
Total Charges	2279.73	1394.55	0	8684.80

5. Churn Distribution

The notebook visualizes churn using count plots and pie charts.

Churn Percentage

Status	Approx. Count	Percentage
Not Churned	5,174	73.46%
Churned	1,869	26.54%

Key Insight

- Roughly **1 out of every 4 customers** leaves the company.

6. Gender Analysis

Visualization

- Countplot of Gender vs Churn

Observation

- Male and Female customers show **very similar churn behavior**.
- Gender is **not a strong predictor** of churn.

7. Senior Citizen Analysis

Findings

- Senior citizens represent a smaller portion of customers.
- However, **their churn percentage is noticeably higher** than non-senior customers.

Insight

- Senior citizens appear to be a **higher-risk customer segment**.

8. Tenure Analysis

Visualization

- Histogram of Tenure by Churn

Findings

- Customers with **short tenure** show the highest churn.
- Customers staying for a long time are more likely to remain loyal.

Insight

- Early customer retention is critical.

9. Contract Type Analysis

Visualization

- Contract vs Churn Countplot

Findings

Contract Type	Churn Risk
Month-to-Month	Highest
One-Year	Moderate
Two-Year	Lowest

Insight

Long-term contracts significantly reduce churn.

10. Service-Based Analysis

The notebook analyzes the following services:

- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

Key Trends

Customers are more likely to churn when they:

 Do not have:

- Online Security
- Online Backup
- Device Protection
- Tech Support

Customers are less likely to churn when they:

 Subscribe to:

- Security services
- Backup services
- Technical support plans

11. Internet Service Impact

Observation

- Fiber Optic customers typically show higher churn than DSL users.
- Customers without internet service generally churn less.

Insight

Internet service quality and pricing may influence customer retention.

12. Payment Method Analysis

Visualization

- Churn by Payment Method

Key Finding

Customers using:

- **Electronic Check** → Highest churn rate

Customers using:

- Automatic bank transfer
- Credit card auto-payment

tend to have lower churn.

Insight

Automatic payment methods are associated with better retention.

Executive Summary

Major Drivers of Churn

Factor	Impact on Churn
Month-to-Month Contracts	 High
Short Tenure	 High
Senior Citizens	 High
Electronic Check Payments	 High
Lack of Online Security	 High
Lack of Tech Support	 High
Lack of Backup/Protection Services	 High
Long-Term Contracts	 Reduce Churn
Auto-Payment Methods	 Reduce Churn

Overall Conclusion

Customer churn rate $\approx$ 26.54%.

The strongest indicators of churn identified in the notebook are:

1. Month-to-month contracts
2. Low tenure customers
3. Senior citizens
4. Electronic check payment users
5. Customers without security, backup, and technical support services

Recommended Retention Actions

- Encourage customers to move to annual contracts.
- Improve onboarding for new customers.
- Offer bundled security/support services.
- Promote automatic payment methods.
- Create targeted retention campaigns for senior citizens and new customers.

Overall Health Score: 73.46% retained customers vs 26.54% churned customers.